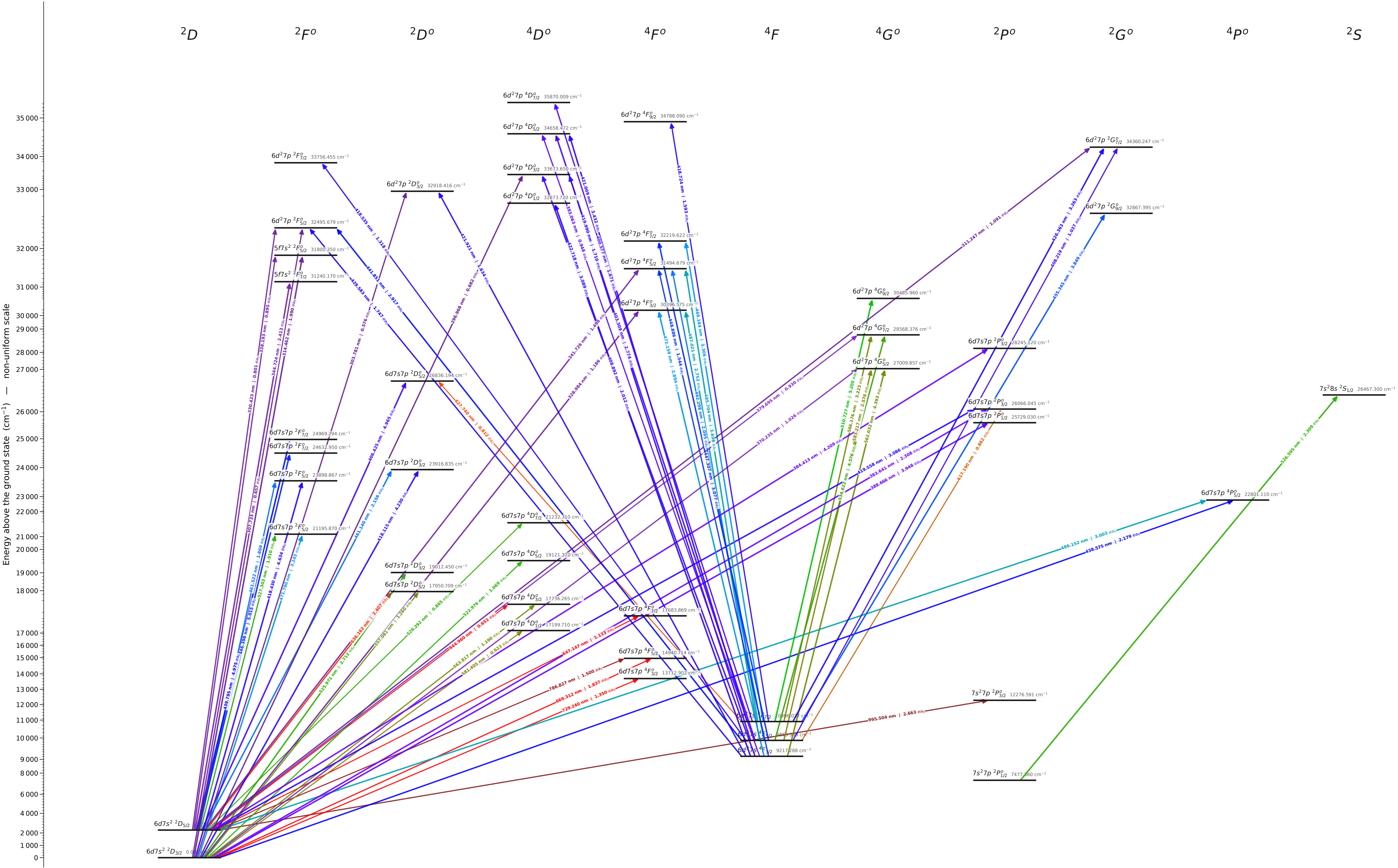
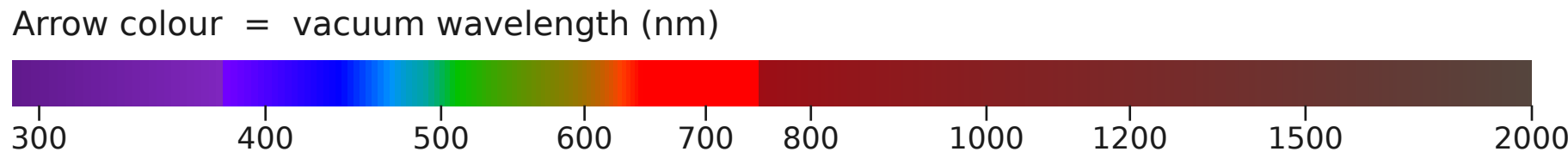


Actinium

Ac (Ac I). Strongest classified transitions below 2 μm (NIST): 45 levels, 70 lines.

Ac⁺ ionisation limit 43394.52 cm⁻¹ = 5.38023 eV (230.444 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $\langle U||er||J' \rangle$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.
Level caption: NIST energy and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\langle U er J' \rangle$ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
6d7s ² ² D _{3/2} - 5f7s ² ² F _{5/2} ^o	314.4619	314.3709	953.3505	1.990	4.300e+07	-	-	-
6d7s ² ² D _{3/2} - 6d7s7p ² P _{3/2} ^o	383.6409	383.5321	781.4404	2.308	4.780e+07	-	-	-
6d7s ² ² D _{5/2} - 6d7s7p ² P _{5/2} ^o	384.4130	384.3040	779.8708	4.209	1.580e+08	-	-	-
6d7s ² ² D _{3/2} - 6d7s7p ² P _{1/2} ^o	388.6660	388.5559	771.3369	3.948	2.690e+08	-	-	-
6d ² 7s ⁴ F _{5/2} - 6d ² 7p ⁴ D _{5/2}	403.3090	403.1951	743.3319	2.774	3.960e+07	-	-	-
6d7s ² ² D _{5/2} - 6d7s7p ² D _{5/2}	406.4254	406.3106	737.6322	4.965	1.240e+08	-	-	-
6d7s ² ² D _{3/2} - 6d7s7p ² D _{3/2}	418.1155	417.9977	717.0087	4.230	1.240e+08	-	-	-
6d7s ² ² D _{3/2} - 6d7s7p ² F _{5/2} ^o	418.4299	418.3120	716.4700	4.634	9.900e+07	-	-	-
6d7s ² ² D _{5/2} - 6d7s7p ² P _{3/2}	419.5579	419.4397	714.5437	3.086	6.530e+07	-	-	-
6d ² 7s ⁴ F _{7/2} - 6d ² 7p ⁴ D _{5/2}	421.0093	420.8907	712.0804	3.432	5.330e+07	-	-	-
6d ² 7s ⁴ F _{3/2} - 6d ² 7p ⁴ D _{3/2}	422.7180	422.5990	709.2020	3.089	1.280e+08	-	-	-
6d7s ² ² D _{5/2} - 6d7s7p ² F _{7/2} ^o	439.7951	439.6716	681.6639	4.975	7.370e+07	-	-	-
6d7s ² ² D _{5/2} - 6d7s7p ² F _{5/2} ^o	446.3983	446.2731	671.5806	5.015	7.160e+07	-	-	-
6d ² 7s ⁴ F _{5/2} - 6d ² 7p ⁴ F _{5/2}	472.1594	472.0274	634.9390	2.894	4.030e+07	-	-	-
6d ² 7s ⁴ F _{7/2} - 6d ² 7p ⁴ F _{5/2}	485.7045	485.5688	617.2323	3.698	4.030e+07	-	-	-
6d ² 7s ⁴ F _{7/2} - 6d ² 7p ⁴ G _{5/2}	510.7270	510.5847	586.9916	5.205	4.120e+07	-	-	-

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.

Γ/2π: natural linewidth of the upper level, 1/(2πτ).